Multi-Role Buffer Protocol

Toxicology Study Material Manufacturing Record

Informal Designation: "GLP Level Batch Record"

Purpose: To document the manufacturing process for non-clinical test articles intended for use in Good Laboratory Practice (GLP) studies. This record provides a complete history of material production, including all raw materials, equipment, process steps, and deviations.

Run: Run-2026-001 Status: ACTIVE Started: Completed:

Project: Organization: Protocol Version: v2 Generated: April 1, 2026

1. General Information

Product Name / Candidate	Multi-Role Buffer Protocol
Batch / Lot Number	Run-2026-001
Target Yield / Scale	
Process SOP / Protocol Reference	Multi-Role Buffer Protocol (v2)

2. Bill of Materials (BOM)

Record all raw materials, media, and buffers utilized in this run. If material data is captured by the system, rows are populated automatically; otherwise complete manually.

Material Description	Part / Lot Number	Required Quantity	Actual Quantity Added	Operator Initials & Date

3. Equipment Log

Record primary equipment used to ensure traceability in the event of equipment failure or contamination.

Equipment Name / Type	Equipment ID (Asset Tag)	Calibration Status (Valid Until)	Operator Initials & Date

4. Execution: Unit Operations

Execute process steps. Unlike GMP, second-person verification (Verifier) is not always strictly mandated for all steps in development/tox manufacturing — apply per your SOP.

4.1 Media Prep

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Weigh Reagents	Weigh NaCl and KCl. NaCl: 8, and KCl: 0.2.	NaCl (g): 8.01	D.S.C.	

			KCl (g): 0.2		
2	Dissolve	Add to 1000 mL water and stir.	1000	D.S.C.	

4.2 QC

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Measure pH	Measure and adjust pH. Target pH: 7.4.	7.38	J.W.	

5. Deviations and Process Comments

Log any deviations from the target parameters or standard operating procedures. In development/tox runs, deviations are expected; the key is rigorous documentation. Run notes are surfaced here.

Step / Status Ref.	Description of Deviation / Observation	Impact Assessment Required? (Y/N)	Lead Reviewer Sign-off
ACTIVE	Observed slight turbidity after buffer prep.		Dr. Sarah Chen 2026-04-01 10:35:00
ACTIVE	[ANOMALY] ANOMALY: Cell viability below threshold at 85%.		Dr. Sarah Chen 2026-04-01 11:20:00

6. Final Disposition & Signatures

By signing below, I certify that the material described herein was produced according to the specified process instructions and that all deviations have been reviewed and assessed for impact on the suitability of the material for its intended non-clinical use.

Wet-Ink Sign-Off

Role	Name (Print)	Signature	Date
Lead Operator / Engineer			
Process Development Lead			
Quality / Compliance (if applicable)			